Seminar in AI - Feature Reuse and Scaling: Understanding Transfer Learning with Protein Language Models

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Abstract

This seminar report critically reviews the work "Feature Reuse and Scaling: Understanding Transfer Learning with Protein Language Models" by Li et al. (2024). The study examines the effectiveness of masked language modeling (MLM) pretraining for protein language models (PLMs) across ten downstream tasks involving structural, global, and local property predictions. By comparing MLM-pretrained PLMs to various baselines, the authors ensure observed benefits are not confounded by overparameterization, inductive biases, or weight statistics transfer. Additionally, the paper explores the downstream task performance scalability across three axes: model size, layer depth, and pretraining performance. The findings reveal that structural prediction tasks benefit from MLM pretraining and scaling, indicating feature reuse, while other tasks either leverage low-level features and do not show scaling behavior, or show no benefits. Strengths of the paper include its well-motivated research question and sound methodology such as the rigorous examination of confounding variables. However, limitations arise from some methodological choices, such as using dataset splits less relevant to current protein engineering challenges. Nevertheless, the study highlights effectively current challenges in protein representation learning. This report is organized into four sections: the paper's story summary, a paper summary, a quality analysis of its content, scientific writing and form, and an overall conclusion.

1 Story summary

1. What is the central question?

   The paper addresses two central questions: (1) Is the masked language modeling (MLM) pretraining objective suitable for pretraining protein language models (PLM) to effectively improve the performance of various downstream tasks within the protein domain? (2) Does the observed performance gain scale with model size, model depth and/or pretraining performance?

2. Why is this question important?

   During pretraining, PLMs can gain additional knowledge from the vast amount of unlabeled protein sequence data available. However, if the pretraining task fails to learn relevant features for downstream tasks such as structure or function prediction, pretraining may be considered a waste of resources. Moreover, if downstream performance does not scale along the examined axes, time and resources can be saved by using smaller pretrained models that yield the same benefit as a larger model.

3. What evidence/data (variables) are needed to answer this question?

Downstream performance results should improve for encoders with better pretraining performance, larger models and greater model depth. Furthermore, a re-initialized encoder or an encoder with permuted weights should perform worse than a pretrained encoder. This would show that improvements in downstream tasks are not caused by overparameterization, inductive biases and/or transferred weight statistics but rather by feature reuse.

4. What methods are used to get this evidence/data?

   Li et al. (2024) used two distinct pretrained PLM architectures as encoders, ESM1 and CARP, and evaluated their sequence embeddings on ten downstream tasks using finetuned linear prediction heads. The scaling potential of the encoders was analyzed by using models of different sizes, by using their hidden layer representations as input embeddings for the prediction head, and by using different model checkpoints collected during pretraining. The performance comparison also included newly initialized models and models with layer-wise permuted weights.

5. What analyses must be applied for the data to answer the central question?

   To demonstrate that embeddings from models with randomly initialized or layer-wise permuted weights perform significantly worse than those from pretrained models, a one-sample t-test must yield a one-tailed p-value < 0.05. Furthermore, to show that each scaling axis is strongly correlated with downstream performance, a Spearman's rank correlation analysis must produce a correlation coefficient > 0.9.

6. What evidence/data (values for the variables) were obtained?

   Performance on the three structural downstream tasks improved with transfer learning (TL) and scaled across all three axes. Five other tasks also showed better performance with pretrained models compared to differently initialized ones, but their performance scaled only along the model size axis—or not at all. The remaining two tasks neither scaled along any axes nor showed improved performance over models with randomly initialized or permuted weights.

7. What were the results of the analyses?

   The analysis showed that structural downstream tasks statistically benefited from TL, with performance being strongly positively correlated across all three axes, suggesting effective use of pretrained features. Five other tasks also statistically benefited from TL, but their performance correlated only with scaling model size because of overparameterization for three out of the five tasks, indicating reliance on low-level features for performance. The two remaining tasks did not statistically benefit from TL and instead appeared to rely on overparameterization and/or inductive bias for any observed performance improvement.

8. How did the analyses answer the central question?

   The analysis showed that only a subset of downstream tasks significantly benefits from MLM pretraining. Furthermore, it demonstrated that downstream performance scales with the examined axes exclusively for structural downstream tasks. Other tasks can benefit from pretraining without exhibiting scaling behavior, because the improvements primarily rely on low-level features learned early in the pretraining process.

9. What does this answer tell us about the broader field?

   The MLM pretraining objective is insufficient for enhancing downstream tasks focused on function prediction, which is essential for protein engineering. Li et al. (2024) highlighted the need to design new pretraining objectives that go beyond learning features solely useful for structural downstream tasks.

10. Did the paper answer the question satisfactorily? Why (not)?

    Yes, Li et al. (2024) answered the question satisfactorily. (1) They demonstrated that, for most downstream tasks, encoders pretrained with MLM yield sequence embeddings with a significant performance improvement over one-hot embeddings, or embeddings produced by randomly-initialized encoders or encoders with permuted weights. (2) The authors showed that only structural downstream tasks benefit from scaling the encoders along all three axes, while other downstream tasks might only scale with model size due to overparameterization.

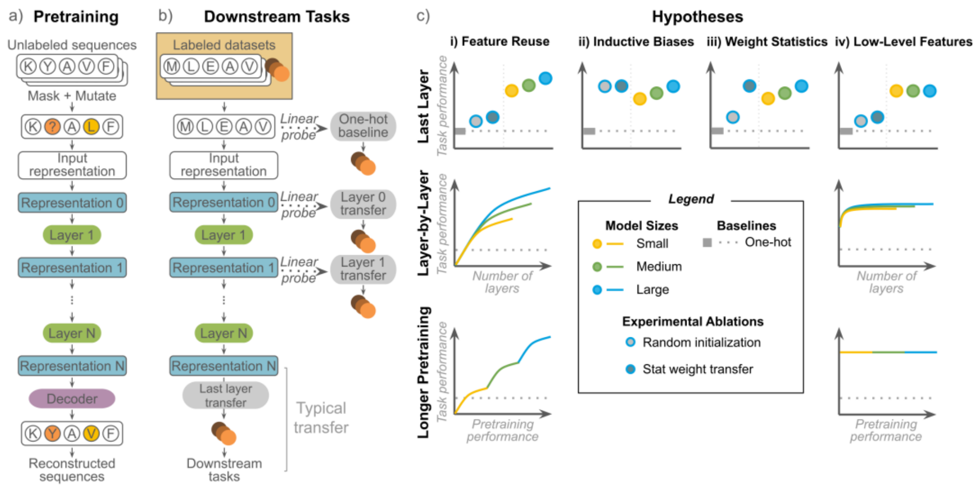

Figure 1: Overview of pretraining and finetuning procedure, and hypothesis analysis. Visualization taken from Li et al. (2024).

2 Paper Summary

This chapter summarizes the paper "Feature Reuse and Scaling: Understanding Transfer Learning with Protein Language Models" by Li et al. (2024), and is structured similarly to a typical machine learning paper. It starts with an introduction in Section 2.1, followed by a method description in Section 2.2, and presents the experimental setup and their respective results in Section 2.3. Finally, the chapter ends with a limitation section (Section 2.4) and a conclusion in Section 2.5.

2.1 Introduction

Proteins form the foundation of life, as they are essential for performing a wide range of critical biological functions, including signaling, enzyme activity, and structural support. These functions are mainly determined by a protein's amino acid sequence. However, the relationship between a protein's function and sequence is poorly understood to this day. Biologically testing these functions in wet labs is both costly and labor-intensive, encouraging the use of computational methods for prediction. Protein Language Models (PLMs) have become a popular framework for improving such computational methods. They combine language models with transfer learning (TL), a two-stage training paradigm. In the first stage, pretraining, models are trained on large datasets with a general-purpose task to learn foundational representations or features. In the second stage, finetuning, the pretrained model is further trained on smaller, task-specific datasets to improve downstream task performance. In the case of PLMs, TL typically involves pretraining on large, unlabeled protein sequence datasets using self-supervised tasks, most commonly masked language modeling (MLM), where the model predicts randomly masked segments of a sequence. The pretrained PLM is then often used as a sequence encoder, or further finetuned along with a prediction head for tasks such as structure or function prediction. Despite the widespread use of PLMs, the reasons why TL with PLMs improves performance in various protein downstream tasks remain poorly understood.

Li et al. (2024) investigated whether MLM is a suitable pretraining objective for a wide range of downstream tasks and whether scaling further improves downstream performance. They address these questions by evaluating three hypotheses: (1) pretraining is effective due to overparameterization and/or inductive biases, (2) solely the statistics of pretrained weights enhance downstream performance, and (3) only low-level features are reused for downstream tasks. To evaluate these hypotheses, the authors conduct a systematic analysis, including 370 experiments, examining the performance of two pretrained PLM architectures across three scaling axes: model size, depth and pretraining time.

2.2 Methods

Training framework. Li et al. (2024) employed two pretrained PLMs: CARP (Yang et al., 2024), based on CNNs, and ESM1 (Rives et al., 2021), which uses a transformer-based architecture. This setup allows for evaluating whether downstream performance improvements are due to specific architecture's inductive bias. Both models were pretrained using the MLM objective on UniRef50 (Suzek et al., 2014) and have publicly available models of comparative sizes, with respect to parameter numbers and embedding dimensions. The PLMs were used as encoders. Li et al. (2024) froze their weights and used their hidden-layer representations as input for the prediction head, which was finetuned for each downstream task. Typically, the last-layer representation was used for the embedding, unless otherwise specified. Linear prediction heads were used for all experiments, specifically a linear classifier trained with Pytorch (Paszke et al., 2019) or a ridge regression model trained with Scikit-learn (Buitinck et al., 2013), depending on the task. In addition to linear probes, Li et al. (2024) also experimented with attention-pooling and using a small neural network as a prediction head. However, this approach performed significantly worse on the downstream tasks with CARP embeddings. Thus, Li et al. (2024) performed all subsequent experiments using linear probing as a prediction head. For tasks requiring a global sequence representation, a mean pooling step over the sequence length was applied before feeding the embedding to the prediction head.

Downstream tasks. Li et al. (2024) evaluated a total of ten downstream tasks, which can be divided into the following categories:

- **Structure prediction:** Three tasks involve residue-level three-class secondary structure prediction (SS3), where each amino acid is classified as part of either an α-helix, a β-sheet or a coil. This task was taken from TAPE (Rao et al., 2019), a popular benchmark for evaluating protein models, and includes three independent test sets: CB513, TS115 and CASP12. Each test set represents a separate downstream task.
- **Global property prediction:** Thermostability and subcellular localization prediction were used for global property prediction. The thermostability task is a regression problem and involves predicting the melting temperature of 48,000 protein sequences across 13 species (Jarzab et al., 2020). Subcellular localization is a ten-class classification problem aimed at predicting the protein's cell compartment within a eukaryotic cell (Almagro Armenteros et al., 2017; Stärk et al., 2021). Both tasks were highlighted in FLIP (Dallago et al., 2021), another popular protein benchmark focused on tasks relevant to protein engineering.
- **Local property prediction:** Li et al. (2024) used the AAV and GB1 datasets from FLIP to evaluate the model's ability to predict the effects of local sequence variation on protein fitness, making these tasks regression problems. The AAV dataset measures how mutations in specific regions affect the sequence fitness, while the GB1 dataset studies the impact on fitness of simultaneous mutations at four fixed positions. FLIP provides pre-split train-test datasets, from which two splits were used for AAV ("two vs. rest" and "one vs. rest") and three splits for GB1 ("sampled", "low vs. high", and "two vs. rest"). The "sampled" split is a random 80-20% train-test split and "low vs. high" trains models on mutated sequences with lower fitness compared to the wild type and tests models on mutated sequences with higher fitness. "One vs. rest" and "two vs. rest" train the model on sequences with one or two mutations, respectively, and test it on sequences with at least two or three mutations, respectively.

Metrics. Accuracy was the primary metric used by Li et al. (2024) for classification tasks. The ROC-AUC was additionally computed, but its results were not reported in the main paper. For regression tasks, the authors used rank-based metrics, as protein engineers are more interested in the relative ranks of mutations than their absolute function. Spearman's rank correlation and Normalized Discounted Cumulative Gain (NDCG) were the main regression metrics, although only Spearman's rank correlation was reported in the main paper.

Alignment. Li et al. (2024) consider a downstream task to be "aligned" with a pretraining task, if the pretrained PLM embeddings (1) outperform the one-hot baseline, (2) surpass embeddings from different PLM initializations (as described in Section 2.3.1), and (3) show performance scaling across all axes mentioned in Section 2.3.2.

	SS3 - CB513	SS3 - TS115	SS3 - CASP12	Subcellular localization	Thermostability	AAV - two vs many	GB1 - low vs high	GB1 - sampled	AAV - one vs many	GB1 - two vs rest
Transfer > One-hot	✓	✓	✓	✓	✓	✓	✓	✓	✓	✗
Transfer > Random init	✓	✓	✓	✓	✓	✓	✓	✓	~	~
Transfer > Stat transfer	✓	✓	✓	✓	✓	✓	✓	✓	~	✓
Scale with PLM sizes	✓	✓	✓	✓	✓	✓	✗	~	✗	✗
Scale with layer depths	✓	✓	✓	✗	✗	✗	✗	✗	✗	✗
Scale with pretrain losses	✓	✓	✓	✗	✗	✗	✗	✗	✗	✗

Figure 2: Result summary of downstream tasks. ✓ means true, ✗ means false and $\sim$ indicates true for one architecture and false for the other. Visualization taken from Li et al. (2024).

2.3 Experiments

This section outlines the experiments conducted by Li et al. (2024), presents their results, and provides a short discussion. A visual summary of the downstream results is shown in Figure 2.

2.3.1 Baseline and initializations.

Li et al. (2024) used three baselines: (1) a one-hot sequence embedding used as input for the prediction head, and embeddings from PLMs with either (2) randomly initialized or (3) layer-wise permuted weights. The one-hot baseline assesses whether TL generally provides performance improvements. The randomly initialized PLM baseline determines if these improvements are due to overparameterization and/or inductive bias of the encoder. The PLM with layer-wise permuted weights, referred to as stat transfer initialization, evaluates whether the improvements stem from learning useful features relevant to the downstream task or from projecting the input into a sensible scale. The authors examined the one-hot baseline once and performed the initialization experiments using three different seeds per architecture and downstream task. They used a one-sample t-test to determine whether TL outperformed the baselines, using a one-tailed p-value threshold of 0.05.

Transfer learning outperforms one-hot baseline in all tasks except GB1—"two vs. rest". This result was expected, as one-hot encoding serves as a standard baseline for PLMs and is a fundamental benchmark for demonstrating the utility of a newly published PLM. The authors hypothesize that the poor performance on the GB1—"two vs. rest" dataset split may be attributed to its challenging nature. With only 400 training samples and an out-of-distribution test set, it is likely difficult for most pretrained models to surpass the one-hot baseline.

Overparameterization does not drive performance for most tasks. Pretrained PLMs consistently outperformed the differently initialized encoders in all tasks, except the AAV—"one vs. many" and GB1—"two vs. rest" splits. This suggests that, for most tasks, TL benefits are not solely due to overparameterization and/or inductive biases but rather due to features learned during pretraining that translate effectively to the downstream tasks.

Stat transfer reduces performance across all GB1 tasks. Interestingly, the stat transfer initialization led to a decrease in performance across all GB1 tasks in comparison to the one-hot and random initialization baselines. The authors attribute this to the nature of the GB1 task, which focuses on interactions among only four mutated positions, making it a highly local task and thus potentially more sensitive to the effects of random weight permutations.

2.3.2 Scaling experiments.

Scaling factors. In their paper, Li et al. (2024) investigated whether TL yields scalable benefits by examining the downstream task performance across the following scaling axes:

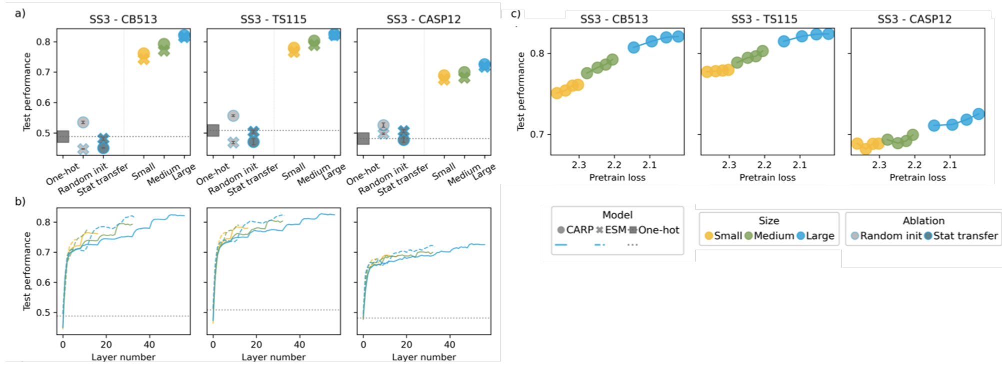

Figure 3: Scaling analysis of SS3 downstream tasks which are aligned with the pretraining task. Visualization adapted from Li et al. (2024).

- **Model size:** Firstly, the authors scaled the size of the pretrained encoders, using a small model with roughly 40 million parameters, a medium model with around 80 million parameters and finally a large model with roughly 645 million parameters.
- **Model depth:** The second scaling axis the authors considered was model depth. For each architecture and model size, they extracted the sequence embeddings from each layer and fed them to the prediction head, which was initialized with the same seed for all layer inputs. A model was considered to scale with layer depth if the Spearman rank correlation between layer number and downstream performance was greater than 0.9. This experiment was performed to assess if only low-level features are used for downstream tasks.
- **Pretraining performance:** The third and final scaling axis is pretraining performance. This was only evaluated for CARP models due to the lack of publicly available ESM1 checkpoints and because the authors did not retrain any PLM. Encoders were considered to scale with pretraining performance if the Spearman rank correlation between negative pretrain loss and downstream task performance was greater than 0.9. The checkpoints were ordered according to their pretraining loss, which was computed using the perplexity of a held-out test set from UniRef50, consisting of 210k sequences that were not used for training CARP.

Structural tasks are aligned with MLM. All SS3 task performances scale across all examined axes, as shown in Figure 3. This suggests that the benefit of using PLM embeddings is not due to overparameterization or inductive biases, but rather because the encoder learns complex, useful features for the prediction head. This finding is consistent with prior research (Rives et al., 2021; Elnaggar et al., 2021; Yang et al., 2024), which showed that secondary structure prediction benefits from using larger or longer pretrained models. Li et al. (2024) conclude that MLM pretraining and scaling up of encoder architectures are effective for secondary structure prediction tasks.

Scaling across model size alone does not indicate alignment. Although TL significantly outperformed the baselines for the subcellular localization, thermostability and AAV—"two vs. many" tasks, performance seems to scale only with model size, as shown in Figure 4. While Li et al. (2024) initially ruled out overparameterization as a contributing factor, they hypothesized that parameterization might still influence performance, which was not apparent in the previous baseline experiments. After rerunning the random initialization experiments with small and medium model sizes, the authors observed scaling across model size also for randomly initialized models. Thus, Li et al. (2024) conclude that scaling model size alone does not indicate task alignment, and additional scaling axes are necessary to establish such alignment.

Many tasks rely on low-level features. The tasks of subcellular localization, thermostability, AAV—"two vs. many", GB1—"low vs. high" and GB1—"sampled" do not show significant scaling behaviour with pretraining performance or layer depth. However, a notable downstream performance increase was observed between the first 3-5 layers across all models, as shown in Figure 4. This suggests that these tasks primarily rely on low-level features, learned early in pretraining. Moreover,

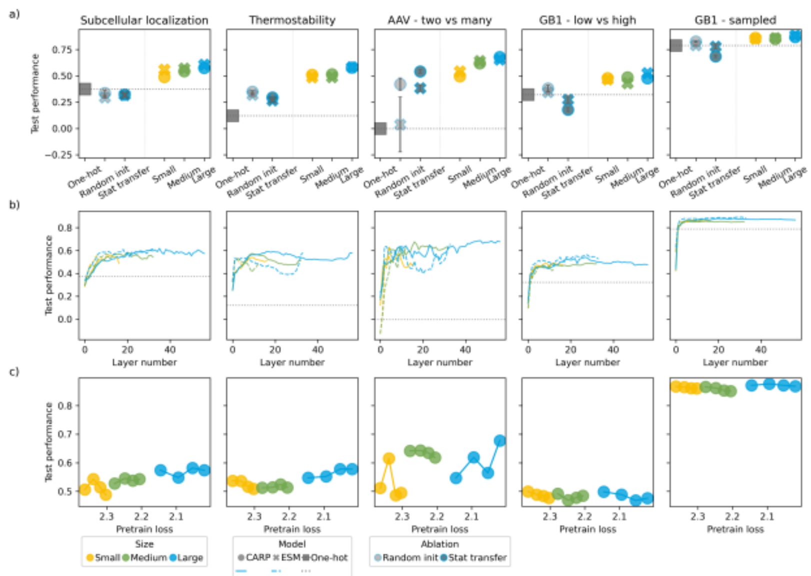

Figure 4: Scaling analysis of downstream tasks that showed significant improvement with transfer learning over baselines, although pretraining and downstream tasks are not aligned. Visualization taken from Li et al. (2024).

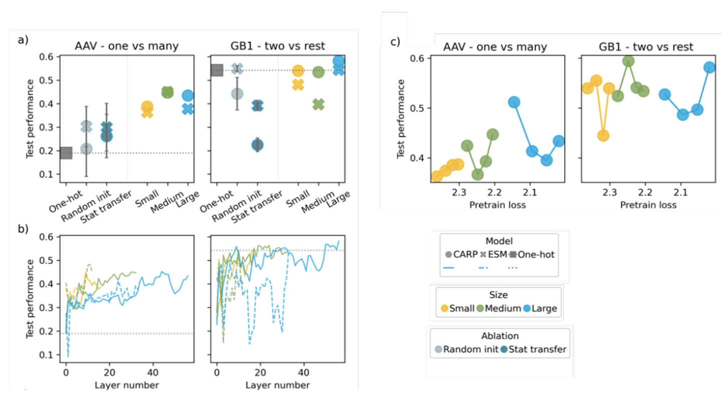

Figure 5: Scaling analysis of downstream tasks that did not show significant improvement with transfer learning over baselines. Visualization adapted from Li et al. (2024).

it indicates that the encoders allocate most of their parameters to encoding structural features, which are mainly relevant for structural prediction tasks.

Some tasks do not benefit from MLM pretraining. The remaining task performances (GB1—"two vs. many" and AAV—"one vs. many") did not scale across any axes, as no TL benefits were observed that are not likely due to overparameterization or inductive bias, as noted in Section 2.3.1. The results are shown in Figure 5.

2.4 Limitations

The authors acknowledge multiple limitations in their work, particularly the inability to retrain pretrained encoders due to a lack of computing resources. This hindered the authors from testing the influence the pretraining dataset had for the TL performance. The choice of pretraining dataset can significantly impact downstream performance, especially when a large distance in domains and/or datasize between pretraining and downstream datasets is apparent, since some downstream data points could be out-of-distribution. This would require the model to re-adjust its weight statistics, which might be hindered by a limited amount of finetuning data.

The authors also highlight a limitation in evaluating only one pretraining objective, specifically MLM. Other objectives, such as autoregressive next-token prediction, have also been proposed for protein TL and might encode additional biological information, potentially influencing the scaling behavior on downstream tasks. However, it remains unclear whether such existing objectives would lead to significantly different results, as they may still overemphasize structural aspects through sequence reconstruction.

Another limitation of the study is its simple finetuning approach that only finetunes the linear prediction head while encoder weights remain frozen. Previous research suggests that end-to-end finetuning would improve overall performance (Rives et al., 2021; Yang et al., 2024). Additionally, more advanced training procedures could further enhance downstream performance and alter the observed alignment patterns. The use of mean pooling, which has been shown to often be suboptimal (Detlefsen et al., 2022), further limits potential performance.

Additionally, the authors note that they could have conducted a more detailed analysis of factors impacting TL. For instance, while Transformer PLMs are known to learn sparse attention matrices, it remains unclear whether downstream performance is driven by sparsity itself or the accurate placement of pairwise attention. This question could not be explored, as they claim their initialization methods do not ensure the preservation of attention matrix sparsity.

Finally, the authors question whether all three scaling axes, particularly model depth, are necessary for detecting task alignment. They note that in computer vision, self-supervised representations learned in a self-supervised manner diverge from those learned in a supervised manner, especially in the final layers. This thus raises doubts about whether a decrease in performance gain across layers truly implies a non-alignment of tasks. Analyzing the model depth was included to explore why downstream performance does not necessarily scale with pretraining performance or model size. The authors conclude that their definition of "alignment" is too simplistic and call for future research to examine the interactions of scaling axes in greater detail.

2.5 Conclusion

The main contribution of the paper is the identification that not all downstream tasks are aligned with the MLM pretraining task and that improved pretraining performance does not imply an improvement in downstream performance. These findings highlight limitations in both the scaling potential of pretrained models and the overall benefits of pretraining itself. Consequently, the authors call for future research to explore new pretraining tasks that are not only aligned with structural downstream tasks but also with functional and property prediction tasks, as they are crucial for progress in protein engineering and the design of novel and optimized proteins.

3 Paper Analysis

This section provides a critical analysis of the paper by Li et al. (2024), reviewing the quality of its content and form, such as layout, writing style and graph interpretability.

3.1 Content Analysis

Research questions are well-motivated and address critical considerations in protein representation learning. The first question—whether MLM is an effective pretraining objective for protein downstream tasks—evaluates the overall utility of TL in reusing these features for downstream tasks. The second question—whether downstream performance scales with model size, layer depth, and pretraining performance—identifies inefficiencies in feature learning, such as a limited reuse of higher-level features. This could potentially motivate the development of alternative, more scalable pretraining objectives and architectures for the protein domain, making this research relevant to current challenges in protein representation learning.

Interpretation of results is clear and well-reasoned. The authors provided a thorough and coherent analysis of their observed results, which is closely aligned with my own interpretation of the data. This aspect is a notable strength of the paper.

Paper demonstrates several strengths in its methodological approach. One such strength is that the authors employed two distinct machine learning architectures (CNNs and Transformers) as encoders to investigate the influence of inductive biases on downstream performance. Additionally, the specific choice of pretrained models (ESM1 and CARP) is appropriate, as both are trained on the same dataset (UniRef50) using the MLM pretraining objective. Another advantage of the paper is their substitution of TAPE's property prediction tasks with those from FLIP, as concerns about the relevance of TAPE tasks specifically for protein engineering were addressed in FLIP (Dallago et al., 2021). Furthermore, the inclusion of commonly applied splits (such as "two vs. many") of the AAV and GB1 datasets adds relevance to current protein engineering challenges. Moreover, the authors used an appropriately chosen t-test to statistically evaluate the benefits of TL over the baselines, although using only three data points to make a statistically significant evaluation is questionable, as the result might be unreliable. Overall, Li et al. (2024) employed a logically sound experimental setup, varying only one variable per experiment and incorporating a diverse range of downstream tasks. However, some content-related issues of this paper were also detected.

Methodology lacked explanations. A significant drawback of the paper is the lack of clear explanations regarding some of their method choices. For instance, the authors included a mainly balanced set of downstream tasks, including structural, global and local property predictions; however, it remains unclear, why no homology task was included in this set of downstream tasks, despite its relevance to protein biology, as no reasoning was offered by the authors. Another instance concerning the downstream tasks is the lack of explanation for the choice of splits for the local property prediction tasks. FLIP authors note that the "sampled" split is less representative of real-world tasks, with "one vs. many" and "two vs. many" representing more relevant data splits (Dallago et al., 2021). While the "two vs. many" split is thus justified for both datasets, the decision to use both the "sampled" and "low vs. high" splits (only) for the GB1 dataset remains unclear. Providing an explanation would clarify this decision, as the current approach seems arbitrary and suggests that the authors may have adjusted the splits progressively until the analysis results fit the desired patterns. Additionally, since results from the GB1 "sampled" and "low vs. high" splits showed identical scaling patterns, and as the "low vs. high" split is more challenging and relevant to protein engineering (Dallago et al., 2021), the inclusion of the "sampled" split seems primarily aimed at increasing the total number of experiments, since the "low vs. high" split would be sufficient to verify the authors' hypotheses. The inclusion of both data splits would be more justifiable if the "sampled" split for the AAV dataset had also been used as a baseline, or if the authors had discussed how training set size and splitting strategy impact downstream performance and scaling behavior for the same underlying task. Another choice in need of explanation is the exclusion of ROC-AUC as a primary metric. Given that subcellular localization is a highly imbalanced task, as noted by Stärk et al. (2021), ROC-AUC would likely be a more robust metric than accuracy, as it is better suited to handle highly imbalanced classification data.

Ideas are not novel but are supported by existing related work. I appreciated that the authors acknowledged in the related work section that their hypotheses are rooted in prior research, particularly from the computer vision and natural language processing domains. While this does not make their approach novel—since they apply existing analysis techniques and explore previously identified issues of TL—their contribution lies in transferring these well-known concepts to the protein domain to identify current limitations of TL in the case of PLMs.

Slight discrepancy in alphabet size and embedding dimensions between CARP and ESM1 could influence the architectures' expressive power. This represents a minor limitation not addressed in the paper, as differences in expressive power between architectures might lead to different analysis patterns than those observed. While it can be argued that such differences are negligible if both architectures perform similarly across finetuning datasets and scaling axes nevertheless, I believe it would have been beneficial to acknowledge this potential limitation, perhaps alongside the mentioned pretraining dataset limitations.

Authors refer to supplemental materials that do not align with the paper's content. For example, they stated that the main metrics (such as accuracy and Spearman's rank correlation) are discussed in the main text, while additional metrics are provided in the supplemental materials. However, the only supplemental materials including a results CSV file is currently solely available with an earlier version of the paper on bioRxiv, and these results do not match those in the final ICML 2024 publication. Consequently, readers must rerun the experiments to verify the paper's figures and examine the additional metrics mentioned by the authors. Moreover, the code generating the results CSV file labels the metrics as "test performance 1" and "test performance 2", making it difficult to identify the metrics without consulting the paper's code. Notably, contrary to intuition, "test performance 1" refers to the secondary metric, such as ROC-AUC, while "test performance 2" corresponds to the primary metric, such as accuracy.

Some statements made in the GitHub README are incorrect. The paper's GitHub README states that the layer-wise embeddings produced by CARP and ESM1 are the most time-consuming process of their experiments. Therefore, the authors claim that these embeddings can be downloaded from their Zenodo repository, which is linked. However, as of 03.12.2024, only pretraining model checkpoints and downstream task data are available, requiring readers to execute the most resource-intensive process, in terms of time and VRAM requirements, themselves.

In conclusion, the paper's content exhibits several strengths, including a relevant research question for protein TL, a sound methodology, and a well-reasoned interpretation of experimental results. However, its main drawback is the lack of explanations for certain methodological choices, such as the choice of data splits for the AAV and GB1 datasets, which may be perceived as suboptimal or partially irrelevant for protein engineering problems, leading to an arguably less qualitative analysis.

3.2 Scientific Writing & Layout Analysis

The paper by Li et al. (2024) is well-structured and provides a clear and intuitive overview of their experiments, objectives, and main contributions. However, while the goal of the paper (analyzing the alignment of MLM pretraining to various downstream tasks) is evident throughout the paper, the abstract does not explicitly mention MLM. Perhaps this was intentional to present their work as a general framework for evaluating pretraining objectives within the protein domain. Nevertheless, MLM would be an important keyword of the study and is missing in the abstract. Furthermore, they may also slightly overstate their contributions regarding the number of experiments, as the study comprises numerous small experiments that require relatively modest computational resources.

At times, the authors' writing was not detailed enough, particularly regarding their used training methods. While technical details are accessible in the code, which is available in their GitHub repository, minimal additions to their main text descriptions would have added substantially more clarity. For instance, while reading the methods and experiment section it remained unclear whether the authors finetuned the encoders or whether they used the pretrained models to produce fixed sequence embeddings. They only explicitly stated that the pretrained models were used as frozen encoders in the limitations section. This key detail could have been introduced much earlier, for example when describing the chosen architectures and prediction heads. As a result, it was occasionally unclear which components were trained by Li et al. (2024) and which were downloaded. Another instance of unclear writing occurs in the limitations section, where Li et al. (2024) state that "[...] stat transfer [does] not guarantee that sparsity in attention matrices is preserved." However, in the experiments section, they describe the stat transfer initialization as a random layer-wise permutation of weights, which theoretically should preserve the sparsity in attention matrices, contrary to their claim.

The figures in the paper, particularly Figure 1c, were highly informative. Figure 1c effectively illustrated the authors' hypotheses and their respective analysis patterns. Overall, the figures were well-chosen, not overwhelming in number, and contributed to a clear and straightforward visualization of the results. The figures include labeled axes and each subpart of the figure is accompanied by detailed descriptions in the figure captions. The color palette used was accessible for individuals with color blindness, while simultaneously ensuring readability in grayscale.

The tables in both the main paper and the appendix were well-structured, easy to read, and clearly captioned. The only minor issue is the omission of an explanation for the "p" column in Table A6 and A8. While it likely refers to the p-value of the Spearman rank correlations, a description in the caption would enhance clarity and completeness.

In conclusion, the paper by Li et al. (2024) is well-structured, with a clear layout that effectively summarizes the contributions and experiments, and supported by useful visualizations and tables. However, it suffers from a lack of detail in certain methodological descriptions.

4 Conclusion

As efficient protein representations are crucial for progress in protein engineering and related fields, Li et al. (2024) investigated whether MLM is a suitable pretraining objective for PLMs and whether scaling model size, depth, and/or pretraining performance improves downstream task performance. The authors found that structural downstream tasks benefit most from MLM pretraining, showing consistent gains across all scaling axes, while other tasks displayed benefits without scaling behavior or no benefits at all, perhaps due to task difficulty. The study's strengths include its well-justified research questions and sound methodology, which carefully controlled for confounding factors like overparameterization or inductive bias. However, certain methodological choices lacked justification, such as dataset splits of the local property tasks and the use of accuracy as a primary metric for an highly imbalanced task. While the authors' hypotheses for TL are not novel, they are grounded in prior research from related fields and the application by Li et al. (2024) of these analyses to the protein domain effectively highlights current limitations in state-of-the-art protein function prediction. This work provides valuable insights into TL with PLMs, their scaling behavior, and the implications for protein representation learning.

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